

Integumentary and Dermatologic Health

Skin barrier, immunity, sensation, thermoregulation, wounds, appendages, imaging, and device engineering

Volume 03J of the Health Atlas Research Suite

Purpose

Integrate skin structure, barrier, immunity, circulation, sensation, appearance, wound repair and systemic signaling with validated measurement and safe device design.

Evidence, guidance, and standards cutoff: 13 August 2026

August 2026

Document status and intended use

This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

Evidence boundary

This volume does not diagnose lesions from photographs, prescribe burn resuscitation, recommend consumer cosmetic procedures or replace urgent evaluation of rapidly progressive rash, infection, necrosis or airway-threatening allergy.

How to use this volume

Skin health is the ability to maintain a resilient barrier, regulate heat and fluid exchange, sense the environment, resist infection, heal injury, and support identity and participation. Visual pattern recognition is powerful but vulnerable to lighting, skin-tone bias, context loss and overconfidence; complete profiling combines history, function, standardized observation, targeted measurement, tissue or microbial reference when needed, and trajectory.

Questions this volume is designed to answer

1. What dimensions define skin health beyond appearance?
2. How do barrier, immunity, microbes, circulation, sensation and environment interact?
3. What can photographs and AI establish, and where is tissue reference required?
4. How should wound and skin-contact sensors be calibrated and validated?
5. How can care avoid skin-tone bias, stigma and overmedicalization?

Scope

- Barrier, dermal, vascular, pigment, immune, sensory, appendage and wound physiology.
- Inflammatory, allergic, infectious, neoplastic, traumatic, ulcerative, pigment, hair/nail and drug-related disease.
- Optical, mechanical, thermal, biochemical, tissue and environmental measurement.
- Topical/systemic therapy, procedures, wound systems, prevention and psychosocial care.
- Engineering of imaging, wearables and wound-contact devices.

Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

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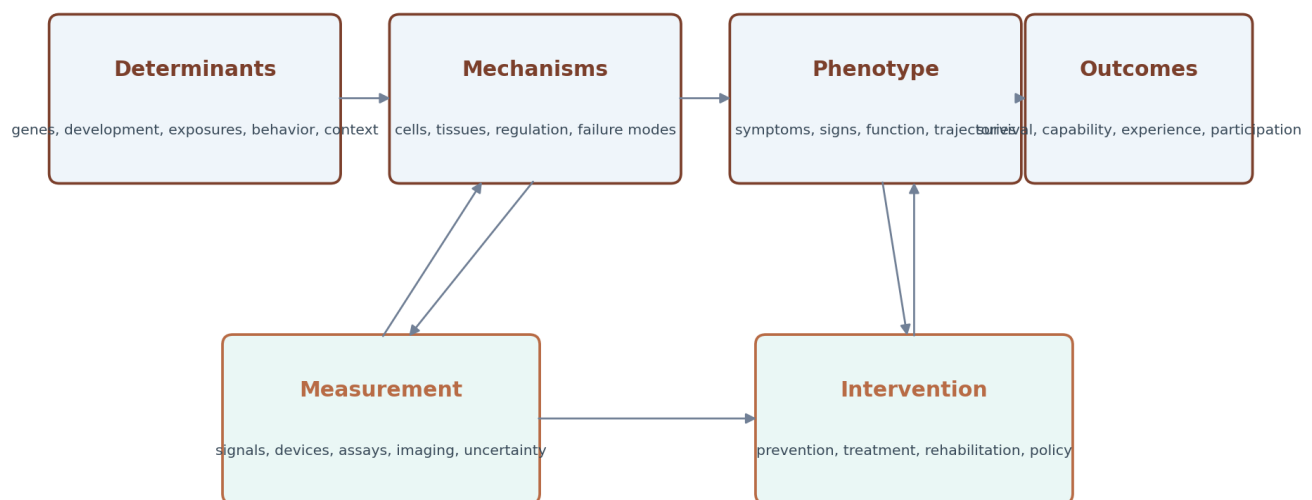
Executive synthesis

Skin health is the ability to maintain a resilient barrier, regulate heat and fluid exchange, sense the environment, resist infection, heal injury, and support identity and participation. Visual pattern recognition is powerful but vulnerable to lighting, skin-tone bias, context loss and overconfidence; complete profiling combines history, function, standardized observation, targeted measurement, tissue or microbial reference when needed, and trajectory.

Central organizing principle

Integrate skin structure, barrier, immunity, circulation, sensation, appearance, wound repair and systemic signaling with validated measurement and safe device design.

Integumentary and Dermatologic Health



A health profile is a causal, temporal, functional, and measurement model - not a single score.

Figure 1.1: The causal and measurement architecture used throughout this volume.

1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.
- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.

- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.
- Integration with other atlas domains rather than treating one specialty as a closed system.

Safety boundary

Rapidly spreading pain, necrosis, mucosal detachment, airway swelling, anaphylaxis, extensive burn, severe blistering, toxic appearance or ischemic/infected wound can require emergency care. Skin-contact and imaging devices require biocompatibility, cleaning, privacy and bias controls.

Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

Period	Development	What it made possible	Residual limitation
Early traditions	Observation of color, lesions, wounds, hair and touch guided early diagnosis and remedies.	Created observational language for skin health and practical care.	Mechanisms and categories mixed empirical findings with culture-specific theory.
Anatomical and pathological era	Microscopy, histology and cellular pathology connected lesion morphology to tissue layers.	Localized structure and tissue change.	Static anatomy poorly represented dynamics, perception, function and context.
Instrumental and laboratory era	Culture, allergy testing, standardized photography, dermoscopy and modern wound care expanded measurement.	Turned previously subjective phenomena into reproducible signals and images.	Thresholds and reference populations sometimes became mistaken for health itself.
Molecular and therapeutic era	Immunology, cancer genetics, retinoids, biologics and barrier science enabled targeted care.	Connected cells, genes, microbes and signaling to targeted prevention and treatment.	Mechanistic specificity often exceeded evidence of patient-important benefit.
Current multimodal era	Confocal/OCT imaging, digital dermatology, spatial biology and smart wound sensing create longitudinal multimodal profiles.	Combines imaging, physiology, function, patient report, longitudinal sensing and computation.	Bias, interoperability, overdiagnosis, access and real-world utility remain limiting.
Systems interpretation	Skin health is now best modeled across tissue state, neural perception, behavior, environment, participation and care.	Supports primary-home and cross-volume reasoning.	Training, billing and datasets still separate closely coupled phenomena.

2.0.1 Observation of color, lesions, wounds, hair and touch guided early diagnosis and remedies.

Scientific consequence. Created observational language for skin health and practical care.

What remained unresolved. Mechanisms and categories mixed empirical findings with culture-specific theory.

Evidence anchors: ^[3,5]

2.0.2 Microscopy, histology and cellular pathology connected lesion morphology to tissue layers.

Scientific consequence. Localized structure and tissue change.

What remained unresolved. Static anatomy poorly represented dynamics, perception, function and context.

Evidence anchors: [3,5]

2.0.3 Culture, allergy testing, standardized photography, dermoscopy and modern wound care expanded measurement.

Scientific consequence. Turned previously subjective phenomena into reproducible signals and images.

What remained unresolved. Thresholds and reference populations sometimes became mistaken for health itself.

Evidence anchors: [3,5]

2.0.4 Immunology, cancer genetics, retinoids, biologics and barrier science enabled targeted care.

Scientific consequence. Connected cells, genes, microbes and signaling to targeted prevention and treatment.

What remained unresolved. Mechanistic specificity often exceeded evidence of patient-important benefit.

Evidence anchors: [3,5]

2.0.5 Confocal/OCT imaging, digital dermatology, spatial biology and smart wound sensing create longitudinal multimodal profiles.

Scientific consequence. Combines imaging, physiology, function, patient report, longitudinal sensing and computation.

What remained unresolved. Bias, interoperability, overdiagnosis, access and real-world utility remain limiting.

Evidence anchors: [3,5]

2.0.6 Skin health is now best modeled across tissue state, neural perception, behavior, environment, participation and care.

Scientific consequence. Supports primary-home and cross-volume reasoning.

What remained unresolved. Training, billing and datasets still separate closely coupled phenomena.

Evidence anchors: [1]

Normal foundations: structure, physiology, and regulation

Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

3.0.1 Epidermal barrier

Core principle. Stratified keratinocytes and lipids limit water loss, chemicals and pathogen entry while renewing continuously.

Control architecture. Differentiation, filaggrin/lipid processing, pH, hydration, microbiome and mechanical integrity.

Failure modes. Dryness, fissure, dermatitis, infection, allergen penetration and excess transepidermal loss.

Observable consequences. Appearance, symptoms, TEWL, hydration, microscopy/biopsy and infection outcomes.

Cross-system coupling. Immune, environmental, fluid/thermal and psychosocial systems.

Evidence anchors: [3,4,5]

3.0.2 Dermal matrix and mechanics

Core principle. Collagen, elastin, extracellular matrix, vessels and fibroblasts provide strength, elasticity and repair.

Control architecture. Genetics, load, age, hormones, UV, inflammation and remodeling enzymes.

Failure modes. Fragility, fibrosis, scarring, wrinkling, ulcers or connective-tissue manifestations.

Observable consequences. Thickness, elasticity, ultrasound/OCT, histology and wound behavior.

Cross-system coupling. Musculoskeletal, vascular, immune and aging systems.

Evidence anchors: [5,8]

3.0.3 Cutaneous circulation and thermoregulation

Core principle. Dense microvascular networks and sweat glands exchange heat and support tissue metabolism.

Control architecture. Autonomic tone, local mediators, core temperature, hydration and environment.

Failure modes. Heat illness, cold injury, flushing, ischemia, pressure injury or impaired healing.

Observable consequences. Temperature, perfusion, sweat, capillary refill, imaging and systemic physiology.

Cross-system coupling. Cardiovascular, autonomic, renal and integrated homeostasis.

Evidence anchors: [2,13]

3.0.4 Pigmentation and photobiology

Core principle. Melanin and cellular repair reduce ultraviolet injury while light also participates in vitamin and circadian biology.

Control architecture. Melanocyte biology, genetics, UV spectrum/dose, DNA repair and immune response.

Failure modes. Sunburn, photoaging, pigment change, carcinogenesis and photosensitivity.

Observable consequences. Colorimetry, UV dosimetry, dermoscopy, histology and molecular damage.

Cross-system coupling. Cancer, endocrine/vitamin, environment and equity in measurement.

Evidence anchors: [3,9,13]

3.0.5 Skin immunity and microbiome

Core principle. Resident immune cells, epithelial signals and microbial communities defend tissue while maintaining tolerance.

Control architecture. Barrier state, antimicrobial peptides, sebum, moisture, immune networks and exposures.

Failure modes. Infection, allergy, autoimmunity, dysbiosis or chronic inflammation.

Observable consequences. Microbiology, immune markers, lesion pattern and treatment response.

Cross-system coupling. Immune, infection, GI and environmental systems.

Evidence anchors: [3,6]

3.0.6 Sensation, itch and pain

Core principle. Cutaneous mechanoreceptors, thermoreceptors, nociceptors and itch pathways encode contact and threat.

Control architecture. Peripheral fibers, mediators, spinal/brain processing, attention and stress.

Failure modes. Neuropathic pain, chronic itch, sensory loss, scratching injury or dysesthesia.

Observable consequences. Sensory exam, quantitative testing, patient report, sleep and function.

Cross-system coupling. Nervous, brain/behavior, sleep and psychosocial systems.

Evidence anchors: [19,20,32]

3.0.7 Hair, nails and glands

Core principle. Appendages provide protection, thermoregulation, lubrication and social/identity signals.

Control architecture. Follicle cycling, androgens, thyroid/nutrition, immune state, trauma and local ecology.

Failure modes. Alopecia, nail dystrophy, acne, hidradenitis or gland dysfunction.

Observable consequences. Pattern, microscopy/dermoscopy, labs when indicated and quality-of-life impact.

Cross-system coupling. Endocrine, immune, nutrition and psychosocial systems.

Evidence anchors: [3,18,31]

3.0.8 Wound healing

Core principle. Hemostasis, inflammation, proliferation, angiogenesis, matrix deposition and remodeling repair skin after injury.

Control architecture. Perfusion, infection, pressure, glucose, nutrition, oxygen, immune state and mechanical environment.

Failure modes. Chronic ulcer, dehiscence, hypertrophic scar, keloid, contracture or recurrent breakdown.

Observable consequences. Dimensions, tissue/edge/exudate, perfusion, infection, imaging and function.

Cross-system coupling. Vascular, metabolic, immune, musculoskeletal and rehabilitation systems.

Evidence anchors: [5,12,17]

3.0.9 Skin as a systemic signal

Core principle. Color, moisture, lesions, edema and vascular patterns can reflect local disease or systemic physiology.

Control architecture. Blood flow, bilirubin/hemoglobin, hormones, inflammation, drugs and organ failure.

Failure modes. Misattribution, delayed systemic diagnosis or false pattern recognition.

Observable consequences. Standardized observation plus targeted systemic assessment.

Cross-system coupling. All organ systems and pharmacology.

Evidence anchors: [5,30]

3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

Relationship	Expression	Interpretation	Limits
Transepidermal flux concept	$J = -D \frac{\Delta C}{\Delta x}$	Water or solute flux depends on effective diffusivity and concentration gradient across a barrier.	Skin is multilayered, nonuniform and affected by convection, sweat and chamber geometry.
Optical absorbance	$A = -\log_{10}(I/I_0)$	A compact representation of wavelength-dependent attenuation by pigments and tissue.	Scattering, geometry, polarization and device processing make skin reflectance more complex than a homogeneous solution.
Pressure	$P = F/A$	Contact load rises as force is concentrated over smaller area, relevant to pressure injury and device interfaces.	Tissue deformation, shear, duration, perfusion and individual tolerance determine injury.

Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Genetic architecture and family history	Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within integumentary and dermatologic health.	Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.	Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.	Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.
Development and life course	Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.	Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.	Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.	Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.
Nutrition and metabolism	Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.	Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.	Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.	Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Physical activity, conditioning, and sedentary behavior	Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.	Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.	Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.	Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.
Sleep, circadian timing, and recovery	Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.	Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.	History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.	Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.
Stress, emotion, and behavior	Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.	Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.	Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.	Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.
Social, economic, cultural, and care conditions	Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.	Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.	Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.	Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.
Environmental and occupational exposures	Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.	Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.	Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.	Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Health care, medication, and iatrogenic effects	Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.	Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.	Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.	Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

4.0.1 Genetic architecture and family history

Mechanistic pathways. Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within integumentary and dermatologic health.

Potentially modifiable components. Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

How to measure it. Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

Causal and equity cautions. Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

4.0.2 Development and life course

Mechanistic pathways. Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

Potentially modifiable components. Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

How to measure it. Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

Causal and equity cautions. Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

4.0.3 Nutrition and metabolism

Mechanistic pathways. Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

Potentially modifiable components. Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

How to measure it. Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

Causal and equity cautions. Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

Evidence anchors: ^[16]

4.0.4 Physical activity, conditioning, and sedentary behavior

Mechanistic pathways. Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

Potentially modifiable components. Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

How to measure it. Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

Causal and equity cautions. Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

4.0.5 Sleep, circadian timing, and recovery

Mechanistic pathways. Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

Potentially modifiable components. Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

How to measure it. History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

Causal and equity cautions. Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

4.0.6 Stress, emotion, and behavior

Mechanistic pathways. Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

Potentially modifiable components. Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

How to measure it. Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

Causal and equity cautions. Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

4.0.7 Social, economic, cultural, and care conditions

Mechanistic pathways. Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

Potentially modifiable components. Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

How to measure it. Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

Causal and equity cautions. Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

Evidence anchors: ^[15]

4.0.8 Environmental and occupational exposures

Mechanistic pathways. Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

Potentially modifiable components. Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

How to measure it. Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

Causal and equity cautions. Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

Evidence anchors: ^[13,14]

4.0.9 Health care, medication, and iatrogenic effects

Mechanistic pathways. Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

Potentially modifiable components. Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

How to measure it. Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

Causal and equity cautions. Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

Evidence anchors: ^[30]

Major disorders, failure modes, and clinically important phenotypes

The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the dermatologic disorder, lesion pattern, barrier failure, wound state, or functional phenotype, but cross-system processes and functional effects are retained.

5.0.1 Atopic dermatitis

Definition and phenotype. Relapsing pruritic inflammatory disease with barrier dysfunction and variable type-2 immune activation.

Pathophysiology. Barrier defects, immune signaling, microbes, itch-scratch cycling and environment interact.

Determinants and modifiers. Genetics, age, climate, irritants/allergens, infection, stress and access.

Assessment. Clinical morphology/distribution and severity/function; testing only for specific alternative or trigger questions.

Intervention logic. Barrier care, topical anti-inflammatory therapy, trigger reduction and systemic/biologic treatment for selected moderate-severe disease.

Known limits and open questions. Biomarkers do not yet define routine treatment for all; heterogeneity and skin-tone recognition matter.

Suite links. 03H immune; 06 psychosocial; 09 environment.

Evidence anchors: [3,7,15]

5.0.2 Contact dermatitis

Definition and phenotype. Irritant injury or delayed hypersensitivity caused by direct exposures.

Pathophysiology. Barrier disruption or antigen-specific T-cell response follows dose, duration and susceptibility.

Determinants and modifiers. Occupation, products, wet work, PPE, atopy and inequitable exposure control.

Assessment. Distribution/exposure history; patch testing for suspected allergic mechanisms.

Intervention logic. Avoid/substitute exposure, barrier protection and anti-inflammatory treatment; workplace controls.

Known limits and open questions. Patch positivity may not prove current relevance and product ingredients change.

Suite links. 09 occupational; 03H immune.

Evidence anchors: [3,7,14]

5.0.3 Psoriasis

Definition and phenotype. Chronic immune-mediated skin disease with plaques and systemic associations including arthritis and cardiometabolic risk.

Pathophysiology. IL-23/Th17 and keratinocyte networks sustain rapid turnover and inflammation.

Determinants and modifiers. Genetics, obesity, smoking, infection, stress, drugs and access.

Assessment. Clinical phenotype, body surface and special sites, symptoms/QOL, arthritis and comorbidity screening.

Intervention logic. Topical, phototherapy and systemic/biologic options matched to severity, comorbidity and goals.

Known limits and open questions. Treatment sequencing and long-term comparative safety continue to evolve.

Suite links. 03H immune; 03I MSK; 03A/03D cardiometabolic.

Evidence anchors: [3,8,16]

5.0.4 Acne and follicular disorders

Definition and phenotype. Inflammation of pilosebaceous units ranging from comedonal acne to scarring follicular disease.

Pathophysiology. Sebum, follicular keratinization, microbes, hormones and immune response interact.

Determinants and modifiers. Age, androgens, products, medications, occlusion, diet signals and stigma.

Assessment. Clinical pattern, severity/scarring and psychosocial impact; endocrine workup only when phenotype suggests it.

Intervention logic. Topical/systemic antimicrobial-sparing regimens, retinoids/hormonal options and scar prevention.

Known limits and open questions. Diet and microbiome effects remain incompletely causal; antibiotic stewardship is essential.

Suite links. 03D endocrine; 06 psychosocial; 08 AMR.

Evidence anchors: [3,10,18]

5.0.5 Urticaria and angioedema

Definition and phenotype. Transient wheals and/or deeper swelling caused by mast-cell mediators or bradykinin pathways.

Pathophysiology. Histamine-driven activation differs from bradykinin-mediated vascular permeability.

Determinants and modifiers. Infection, drugs, foods, physical triggers, autoimmunity and inherited pathways.

Assessment. Time course, airway/systemic features, trigger pattern and targeted tests; avoid indiscriminate panels.

Intervention logic. Antihistamine-based and selected biologic therapy for mast-cell disease; mechanism-specific therapy for bradykinin states; emergency care for airway/anaphylaxis.

Known limits and open questions. Most chronic cases lack one removable cause; angioedema mechanism can be misclassified.

Suite links. 03H immune; 03B airway; 14 pharmacology.

Evidence anchors: [3,7]

5.0.6 Bacterial, fungal, viral and parasitic skin infection

Definition and phenotype. Infection ranging from superficial localized disease to necrotizing or disseminated illness.

Pathophysiology. Barrier breach, pathogen virulence, local ecology and host defense determine invasion.

Determinants and modifiers. Wounds, diabetes, immune suppression, crowding, climate, travel and AMR.

Assessment. Clinical pattern plus microscopy/culture/PCR or biopsy when severity, atypia or resistance warrants.

Intervention logic. Local/systemic antimicrobial or procedural source control matched to syndrome and susceptibility.

Known limits and open questions. Colonization and contamination complicate tests; overuse drives resistance.

Suite links. 08 infection; 03H immune; 03D diabetes.

Evidence anchors: [4,10,11]

5.0.7 Skin cancer and precancer

Definition and phenotype. Keratinocyte cancers, melanoma and other tumors arising from accumulated genomic and immune changes.

Pathophysiology. UV and other mutagenic injury plus clonal selection, tissue context and immune escape.

Determinants and modifiers. UV, phenotype/genetics, age, immune suppression, prior radiation/arsenic and access to detection.

Assessment. Full clinical/dermoscopic evaluation and histopathology; stage with imaging/molecular tests by indication.

Intervention logic. Sun protection, excision or destructive/radiation/systemic therapies by type and stage.

Known limits and open questions. Population screening net benefit and AI triage transportability remain context-dependent.

Suite links. 14 oncology; 09 UV exposure; 13 AI validation.

Evidence anchors: [3,9,29]

5.0.8 Burns and thermal/chemical/electrical injury

Definition and phenotype. Tissue damage with local depth and possible systemic fluid, inflammatory, respiratory and metabolic consequences.

Pathophysiology. Energy or chemical injury causes necrosis, barrier loss, capillary leak and infection risk.

Determinants and modifiers. Home/work hazards, violence, infrastructure, age, disability and access to burn systems.

Assessment. Immediate airway/systemic assessment, mechanism, extent/depth, associated trauma and serial wound evaluation.

Intervention logic. Remove source, resuscitation and temperature/pain/infection management, wound closure and rehabilitation by severity.

Known limits and open questions. Depth evolves and visual estimates vary; electrical injury can be deeper than skin findings.

Suite links. 03O homeostasis; 09 injury; 10 rehab; 12 trauma systems.

Evidence anchors: [12,14,30]

5.0.9 Chronic wounds and pressure injury

Definition and phenotype. Failure of skin repair from pressure/shear, ischemia, neuropathy, venous disease, metabolic disease or repeated trauma.

Pathophysiology. Persistent injury, poor perfusion, inflammation, infection/biofilm and impaired cellular repair maintain nonhealing.

Determinants and modifiers. Immobility, diabetes, vascular disease, nutrition, poverty, device fit and care fragmentation.

Assessment. Cause-first evaluation: pressure, perfusion, sensation, edema, infection, wound morphology and function.

Intervention logic. Offloading/pressure redistribution, vascular/edema treatment, debridement and moisture balance, infection control and nutrition/system support.

Known limits and open questions. Many advanced products have limited comparative evidence; surface appearance alone misses cause.

Suite links. 03A vascular; 03D diabetes; 03C neuropathy; 10 rehab.

Evidence anchors: [12,17,30]

5.0.10 Autoimmune blistering and connective-tissue skin disease

Definition and phenotype. Immune attack on adhesion structures, vessels or matrix produces blistering, rash, ulceration and systemic clues.

Pathophysiology. Autoantibodies, immune complexes or cellular inflammation target specific skin structures.

Determinants and modifiers. Autoimmunity, age, drugs, malignancy and systemic disease.

Assessment. Morphology, biopsy with immunofluorescence, serology and organ assessment.

Intervention logic. Wound/barrier care and disease-specific immunomodulation with infection/toxicity monitoring.

Known limits and open questions. Serology and disease activity can disagree; broad rashes have many mimics.

Suite links. 03H immune; 03I connective tissue; organ volumes.

Evidence anchors: [3,8]

5.0.11 Pigment disorders

Definition and phenotype. Localized or generalized loss, gain or redistribution of pigment from genetic, immune, inflammatory, hormonal, drug or injury mechanisms.

Pathophysiology. Melanocyte number/function, melanin synthesis/transfer and inflammation alter optical appearance.

Determinants and modifiers. Genetics, UV, inflammation, endocrine disease, drugs and social meaning.

Assessment. Clinical/dermoscopic pattern, Wood lamp or biopsy/labs when indicated, and psychosocial impact.

Intervention logic. Cause-specific anti-inflammatory, light, camouflage or procedural approaches with photoprotection.

Known limits and open questions. Color measurement and recognition are device/illumination/skin-tone sensitive.

Suite links. 03D endocrine; 06 psychosocial; 13 measurement equity.

Evidence anchors: [3,15]

5.0.12 Hair and nail disorders

Definition and phenotype. Changes in growth, structure or loss caused by local disease, systemic illness, nutrition, hormones, infection, trauma or autoimmunity.

Pathophysiology. Cycle disruption, matrix injury, follicular miniaturization/scarring or nail-unit pathology.

Determinants and modifiers. Age, genetics, hormones, illness, drugs, styling/work exposures and nutrition.

Assessment. Pattern, pull/dermoscopy/microscopy/culture and targeted systemic testing.

Intervention logic. Treat cause, protect structure and use topical/systemic/procedural therapy where evidence supports.

Known limits and open questions. Cosmetic marketing exceeds evidence and normal variation is often medicalized.

Suite links. 03D endocrine; 03H immune; 06 identity/QOL.

Evidence anchors: [3,31]

5.0.13 Drug eruptions and severe cutaneous adverse reactions

Definition and phenotype. Medication-related skin reactions ranging from benign exanthem to life-threatening epidermal necrosis or systemic hypersensitivity.

Pathophysiology. Immune or pharmacologic mechanisms vary by latency, drug and host genetics.

Determinants and modifiers. New/recent drugs, dose, infection, immune state, genotype and polypharmacy.

Assessment. Timeline, morphology, mucosa/systemic organs, labs and biopsy selectively; identify alternatives and culprit probability.

Intervention logic. Stop suspected culprit, emergency supportive/specialist care for severe forms, precise documentation and future avoidance.

Known limits and open questions. Causality is difficult with multiple drugs and tests are limited.

Suite links. 14 pharmacology; 12 medication safety; 15 genomics/equity.

Evidence anchors: [30]

5.0.14 Psychodermatologic and body-focused repetitive conditions

Definition and phenotype. Skin symptoms or injury are influenced by stress, perception, compulsive behavior, delusional belief or appearance distress without implying symptoms are unreal.

Pathophysiology. Brain-skin-immune signaling, attention, habit, itch/pain and social meaning interact.

Determinants and modifiers. Stress/trauma, anxiety/OCD-spectrum symptoms, stigma, sleep and health-care experiences.

Assessment. Respectful skin and mental-health assessment, infection/medical exclusion, function and risk.

Intervention logic. Integrated dermatologic, behavioral/psychological and pharmacologic care matched to mechanism.

Known limits and open questions. Fragmented services create underdiagnosis and invalidation.

Suite links. 05 brain/behavior; 06 psychosocial; 01 whole-person.

Evidence anchors: ^[3,21,22]

Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

Stage	Purpose	Components	Output	Common failure
1. Question and urgency	Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.	Chief concern and goals; red flags; prior probability; relevant context and current integumentary and dermatologic health risks.	Urgent action, focused differential, or a prespecified measurement plan.	Testing without a question creates incidental findings and false precision.
2. History and lived phenotype	Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.	Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.	A longitudinal problem representation with hypotheses and missing information.	Administrative codes and checklists can erase chronology, context, and patient priorities.
3. Examination and basic measurements	Test hypotheses using standardized observation and low-burden validated measures.	Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.	Localization, severity estimate, safety screen, and indication for further testing.	Nonstandard technique and unblinded interpretation create systematic error.
4. Targeted laboratory, imaging, and physiology	Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.	Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.	Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.	Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.
5. Function and challenge testing	Measure reserve, adaptation, and real-world impact not visible at rest.	Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.	Capacity, symptom coupling, recovery, and safety limits.	Challenge protocols may not match daily demands and can be unsafe without screening.

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Stage	Purpose	Components	Output	Common failure
6. Longitudinal integration	Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.	Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.	Trajectory, response, relapse, and need for escalation or de-escalation.	Regression to the mean and changing devices can masquerade as improvement or decline.

6.0.1 1. Question and urgency

Purpose. Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

Core components. Chief concern and goals; red flags; prior probability; relevant context and current integumentary and dermatologic health risks.

Interpretation. Urgent action, focused differential, or a prespecified measurement plan.

Failure modes. Testing without a question creates incidental findings and false precision.

6.0.2 2. History and lived phenotype

Purpose. Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

Core components. Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

Interpretation. A longitudinal problem representation with hypotheses and missing information.

Failure modes. Administrative codes and checklists can erase chronology, context, and patient priorities.

Evidence anchors: ^[32]

6.0.3 3. Examination and basic measurements

Purpose. Test hypotheses using standardized observation and low-burden validated measures.

Core components. Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

Interpretation. Localization, severity estimate, safety screen, and indication for further testing.

Failure modes. Nonstandard technique and unblinded interpretation create systematic error.

Evidence anchors: ^[5]

6.0.4 4. Targeted laboratory, imaging, and physiology

Purpose. Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

Core components. Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

Interpretation. Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

Failure modes. Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

6.0.5 5. Function and challenge testing

Purpose. Measure reserve, adaptation, and real-world impact not visible at rest.

Core components. Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

Interpretation. Capacity, symptom coupling, recovery, and safety limits.

Failure modes. Challenge protocols may not match daily demands and can be unsafe without screening.

6.0.6 6. Longitudinal integration

Purpose. Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

Core components. Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

Interpretation. Trajectory, response, relapse, and need for escalation or de-escalation.

Failure modes. Regression to the mean and changing devices can masquerade as improvement or decline.

Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

Measurement is an inference chain



Validation must test every link, including population, setting, motion, missingness, drift, and downstream use.

Figure 7.1: Generic inference chain from latent biology to a clinical or research claim.

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Standardized visual and tactile examination	Noninvasive	Human observation/palpation under controlled illumination with morphology and distribution.	Lesion class, color, scale, border, texture, temperature, tenderness and pattern.	Fast whole-surface contextual assessment.	Training, skin-tone bias, lighting, language and low specificity.
Clinical photography and colorimetry	Noninvasive	Calibrated visible-light image or spectrophotometric reflectance.	Area, color, texture and longitudinal change.	Reproducible documentation and remote review.	Lighting, white balance, distance, privacy and algorithm drift.
Dermoscopy	Noninvasive	Magnified polarized/nonpolarized optical imaging that reduces surface reflection.	Pigment and vascular microstructures supporting lesion triage.	Improves expert assessment of many tumors/inflammatory conditions.	Training, image quality and context; not histology.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Reflectance confocal microscopy and optical coherence tomography	Noninvasive optical imaging	Confocal backscatter or interferometric depth-resolved reflectance.	Cellular/subcellular en face or cross-sectional skin structure.	Near-real-time microanatomy without excision.	Depth, motion, field size, operator and interpretation limits.
High-frequency ultrasound and elastography	Noninvasive	Acoustic reflection and tissue deformation/shear-wave response.	Thickness, fluid, masses, vessels and relative stiffness.	Deeper structural and dynamic information.	Resolution-depth tradeoff, anisotropy and vendor dependence.
Transepidermal water loss	Noninvasive	Humidity gradient or closed-chamber vapor accumulation above skin.	Barrier water-loss rate.	Quantitative barrier proxy.	Environment, acclimation, sweat, site and chamber design strongly affect values.
Corneometry, sebumetry and skin pH	Noninvasive	Capacitance/conductance, lipid collection/optics and electrochemical pH.	Hydration, surface lipid and acidity proxies.	Fast and low burden.	Superficial, product/environment sensitive and device-specific.
Thermography and perfusion imaging	Noninvasive	Infrared emission, laser Doppler/speckle or optical perfusion contrast.	Surface temperature and relative microvascular flow.	Maps inflammation, ischemia and recovery dynamics.	Ambient temperature, emissivity, motion and nonspecific physiology.
Biopsy, histopathology and immunofluorescence	Invasive	Tissue architecture, stains, immune deposits, microbiology and molecular tests.	Definitive pattern for tumors, inflammatory, blistering and infectious disease.	Direct tissue context.	Sampling, scar, processing and lesion-selection error.
Microbiology and molecular pathogen testing	Minimally invasive/invasive sample	Microscopy, culture, antigen, PCR or sequencing.	Pathogen and susceptibility evidence.	Actionable for atypical, severe or resistant infection.	Colonization/contamination and prior treatment.
Patch, prick and provocation testing	In vivo, variable risk	Controlled allergen exposure to skin or supervised challenge.	Delayed hypersensitivity, immediate sensitization or clinical reactivity.	Mechanism support when history is specific.	False relevance, severe reaction and extract variability.
Wound geometry and tissue sensors	Noninvasive or minimally invasive	Photography/3D depth, impedance, pressure, temperature, moisture, oxygen and biochemical sensing.	Area/volume, edge change, load, perfusion/infection proxies and trend.	Potential objective longitudinal monitoring.	Contact contamination, drift, exudate effects and surrogate-only claims.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
UV and environmental exposure dosimetry	Noninvasive wear-able/environmental	Photodiodes/color-changing materials and temperature/humidity/pollutant sensors.	Personal/area exposure dose and context.	Connects prevention to actual exposure.	Body placement, behavior, spectrum, calibration and adherence.
AI-assisted image triage	Noninvasive software	Computer vision over photographs/dermoscopy with metadata.	Risk score, segmentation or differential support.	Potential access and workflow support.	Dataset shift, skin-tone/site bias, shortcut learning, privacy and overreferral.

7.0.1 Standardized visual and tactile examination

Measured quantity and transducer. Human observation/palpation under controlled illumination with morphology and distribution.

Typical outputs. Lesion class, color, scale, border, texture, temperature, tenderness and pattern.

Strengths. Fast whole-surface contextual assessment.

Limitations and artifacts. Training, skin-tone bias, lighting, language and low specificity.

Reference and validation strategy. Image standards, inter-rater studies, pathology/clinical outcome correlation and diverse training.

Clinical/research maturity. Established core.

Evidence anchors: [3,15]

7.0.2 Clinical photography and colorimetry

Measured quantity and transducer. Calibrated visible-light image or spectrophotometric reflectance.

Typical outputs. Area, color, texture and longitudinal change.

Strengths. Reproducible documentation and remote review.

Limitations and artifacts. Lighting, white balance, distance, privacy and algorithm drift.

Reference and validation strategy. Color target/geometry controls, repeatability and diverse-skin validation.

Clinical/research maturity. Established to emerging.

Evidence anchors: [23,28]

7.0.3 Dermoscopy

Measured quantity and transducer. Magnified polarized/nonpolarized optical imaging that reduces surface reflection.

Typical outputs. Pigment and vascular microstructures supporting lesion triage.

Strengths. Improves expert assessment of many tumors/inflammatory conditions.

Limitations and artifacts. Training, image quality and context; not histology.

Reference and validation strategy. Expert/pathology comparison and prospective management utility.

Clinical/research maturity. Established.

Evidence anchors: [3,9]

7.0.4 Reflectance confocal microscopy and optical coherence tomography

Measured quantity and transducer. Confocal backscatter or interferometric depth-resolved reflectance.

Typical outputs. Cellular/subcellular en face or cross-sectional skin structure.

Strengths. Near-real-time microanatomy without excision.

Limitations and artifacts. Depth, motion, field size, operator and interpretation limits.

Reference and validation strategy. Phantoms, histology correlation and clinical utility.

Clinical/research maturity. Advanced/emerging.

Evidence anchors: [9,23]

7.0.5 High-frequency ultrasound and elastography

Measured quantity and transducer. Acoustic reflection and tissue deformation/shear-wave response.

Typical outputs. Thickness, fluid, masses, vessels and relative stiffness.

Strengths. Deeper structural and dynamic information.

Limitations and artifacts. Resolution-depth tradeoff, anisotropy and vendor dependence.

Reference and validation strategy. Tissue phantoms, repeatability and pathology/imaging reference.

Clinical/research maturity. Established to emerging.

Evidence anchors: [23,25]

7.0.6 Transepidermal water loss

Measured quantity and transducer. Humidity gradient or closed-chamber vapor accumulation above skin.

Typical outputs. Barrier water-loss rate.

Strengths. Quantitative barrier proxy.

Limitations and artifacts. Environment, acclimation, sweat, site and chamber design strongly affect values.

Reference and validation strategy. Controlled climate, site protocol, calibration and intervention responsiveness.

Clinical/research maturity. Established research/selected clinical.

Evidence anchors: [3,25]

7.0.7 Corneometry, sebumetry and skin pH

Measured quantity and transducer. Capacitance/conductance, lipid collection/optics and electrochemical pH.

Typical outputs. Hydration, surface lipid and acidity proxies.

Strengths. Fast and low burden.

Limitations and artifacts. Superficial, product/environment sensitive and device-specific.

Reference and validation strategy. Standardized washing/acclimation, repeatability and reference materials.

Clinical/research maturity. Research/cosmetic; limited diagnostic use.

Evidence anchors: [23]

7.0.8 Thermography and perfusion imaging

Measured quantity and transducer. Infrared emission, laser Doppler/speckle or optical perfusion contrast.

Typical outputs. Surface temperature and relative microvascular flow.

Strengths. Maps inflammation, ischemia and recovery dynamics.

Limitations and artifacts. Ambient temperature, emissivity, motion and nonspecific physiology.

Reference and validation strategy. Blackbody/flow phantoms, acclimation and clinical reference studies.

Clinical/research maturity. Emerging.

Evidence anchors: [2]

7.0.9 Biopsy, histopathology and immunofluorescence

Measured quantity and transducer. Tissue architecture, stains, immune deposits, microbiology and molecular tests.

Typical outputs. Definitive pattern for tumors, inflammatory, blistering and infectious disease.

Strengths. Direct tissue context.

Limitations and artifacts. Sampling, scar, processing and lesion-selection error.

Reference and validation strategy. Adequate site/depth, expert integrated pathology and clinicopathologic correlation.

Clinical/research maturity. Established reference.

Evidence anchors: [5,9]

7.0.10 Microbiology and molecular pathogen testing

Measured quantity and transducer. Microscopy, culture, antigen, PCR or sequencing.

Typical outputs. Pathogen and susceptibility evidence.

Strengths. Actionable for atypical, severe or resistant infection.

Limitations and artifacts. Colonization/contamination and prior treatment.

Reference and validation strategy. Specimen-quality controls and clinical adjudication.

Clinical/research maturity. Established to emerging.

Evidence anchors: [10,11]

7.0.11 Patch, prick and provocation testing

Measured quantity and transducer. Controlled allergen exposure to skin or supervised challenge.

Typical outputs. Delayed hypersensitivity, immediate sensitization or clinical reactivity.

Strengths. Mechanism support when history is specific.

Limitations and artifacts. False relevance, severe reaction and extract variability.

Reference and validation strategy. Controls, trained interpretation and outcome correlation.

Clinical/research maturity. Established specialist.

Evidence anchors: ^[3,7]

7.0.12 Wound geometry and tissue sensors

Measured quantity and transducer. Photography/3D depth, impedance, pressure, temperature, moisture, oxygen and biochemical sensing.

Typical outputs. Area/volume, edge change, load, perfusion/infection proxies and trend.

Strengths. Potential objective longitudinal monitoring.

Limitations and artifacts. Contact contamination, drift, exudate effects and surrogate-only claims.

Reference and validation strategy. Geometry phantoms, biocompatibility, reference perfusion/microbiology and utility trials.

Clinical/research maturity. Emerging/frontier.

Evidence anchors: ^[23,26,27]

7.0.13 UV and environmental exposure dosimetry

Measured quantity and transducer. Photodiodes/color-changing materials plus temperature/humidity/pollutant sensors.

Typical outputs. Personal/area exposure dose and context.

Strengths. Connects prevention to actual exposure.

Limitations and artifacts. Body placement, behavior, spectrum, calibration and adherence.

Reference and validation strategy. Traceable source calibration and personal-behavior validation.

Clinical/research maturity. Emerging.

Evidence anchors: ^[13]

7.0.14 AI-assisted image triage

Measured quantity and transducer. Computer vision over photographs/dermoscopy with metadata.

Typical outputs. Risk score, segmentation or differential support.

Strengths. Potential access and workflow support.

Limitations and artifacts. Dataset shift, skin-tone/site bias, shortcut learning, privacy and overreferral.

Reference and validation strategy. Locked external prospective validation, subgroup/calibration, clinician workflow and outcome utility.

Clinical/research maturity. Emerging; not a universal autonomous diagnosis.

Evidence anchors: ^[24,29]

Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

Subsystem / design issue	Requirements	Dominant risks	Verification
Intended use and measurand	Define the precise integumentary and dermatologic health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.	A broad wellness claim can hide several incompatible clinical claims.	Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.
Sensor and transduction	Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.	A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.	Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.
Timing and synchronization	Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.	Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.	Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.
Analog, acquisition, and digital integrity	Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.	Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.	Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.
Calibration and traceability	Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.	Model calibration to one device lot, site, or operator may not transport.	Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.
Artifact and quality channels	Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.	Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.	Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

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Subsystem / design issue	Requirements	Dominant risks	Verification
Algorithms and uncertainty	Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.	Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.	Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.
Safety, usability, cybersecurity, and privacy	Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.	Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.	Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

8.0.1 Intended use and measurand

Design requirements. Define the precise integumentary and dermatologic health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

Failure modes. A broad wellness claim can hide several incompatible clinical claims.

Bench verification. Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

Human validation boundary. Do not enroll people until the measurement purpose and risk controls are reviewable.

Evidence anchors: [23,26]

8.0.2 Sensor and transduction

Design requirements. Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

Failure modes. A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

Bench verification. Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

Human validation boundary. Bench testing should precede contact with participants.

Evidence anchors: [26,27]

8.0.3 Timing and synchronization

Design requirements. Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

Failure modes. Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

Bench verification. Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

Human validation boundary. Store both device time and host/reference time.

8.0.4 Analog, acquisition, and digital integrity

Design requirements. Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

Failure modes. Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

Bench verification. Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

Human validation boundary. Preserve raw or minimally processed channels where ethically and technically possible.

Evidence anchors: ^[25]

8.0.5 Calibration and traceability

Design requirements. Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

Failure modes. Model calibration to one device lot, site, or operator may not transport.

Bench verification. Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

Human validation boundary. Calibration data and version must travel with each observation.

Evidence anchors: ^[25]

8.0.6 Artifact and quality channels

Design requirements. Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

Failure modes. Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

Bench verification. Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Human validation boundary. Quality rejection must not silently exclude populations.

Evidence anchors: [29]

8.0.7 Algorithms and uncertainty

Design requirements. Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

Failure modes. Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

Bench verification. Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

Human validation boundary. A model should abstain when inputs are outside validated conditions.

Evidence anchors: [24,29]

8.0.8 Safety, usability, cybersecurity, and privacy

Design requirements. Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

Failure modes. Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

Bench verification. Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

Human validation boundary. Research exemptions do not remove the ethical duty to minimize foreseeable harm.

Evidence anchors: [27]

Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Barrier care and irritant reduction	Prevention/treatment	Restore moisture/lipids, reduce friction/wet work and avoid damaging exposures.	Foundational for dermatitis, xerosis and wound prevention.	Cost, contact allergy, adherence and occupational infeasibility.	Symptoms, barrier measures, flares, skin integrity and exposure.
Topical pharmacotherapy	Local treatment	Deliver anti-inflammatory, antimicrobial, keratolytic, retinoid or other agents to skin.	Strong for many localized conditions when vehicle and adherence fit.	Irritation, atrophy/systemic absorption, resistance and contact allergy.	Lesion response, adverse effects, amount/use and escalation need.
Photoprotection and therapeutic phototherapy	Prevention/treatment	Reduce harmful UV exposure or deliver controlled wavelengths for selected inflammatory/pigment disease.	Sun protection reduces UV injury; phototherapy effective for selected diseases.	Vitamin/behavior concerns are often overstated; burns, photoaging and logistical burden.	Exposure, lesion response, cumulative dose and adverse effects.
Systemic and biologic dermatologic therapy	Disease modification	Modulate immune, hormonal, microbial or keratinization pathways.	Major benefit in selected severe psoriasis, dermatitis, acne and other disorders.	Infection, organ/reproductive effects, monitoring and cost.	Disease target, labs/organ safety, infection, QOL and adherence.
Antimicrobial stewardship and source control	Treatment/systemic	Use appropriate topical/systemic agents, drainage/debridement and hygiene.	Effective when infection is correctly identified and source controlled.	Resistance, allergy, microbiome injury and unnecessary treatment.	Culture/susceptibility, response, recurrence and antimicrobial use.

Continued on next page

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Wound offloading, compression and perfusion treatment	Cause-directed rehabilitation	Remove pressure/shear, control edema and restore blood flow where possible.	Often more important than advanced dressings for chronic wounds.	Ischemia risk with compression, adherence, device injury and access.	Pressure/load, perfusion, edema, wound trajectory, infection and function.
Surgery and procedural dermatology	Definitive/local	Biopsy, excision, Mohs surgery, drainage, laser or reconstructive procedures.	Strong for many tumors and selected structural/inflammatory problems.	Scar, infection, pigment change, recurrence and access.	Margins/pathology, healing, recurrence, function and cosmetic goals.
Emergency burn and severe-reaction care	Acute/system	Stabilize airway/circulation, stop exposure, control temperature/pain, remove culprit and provide specialized wound/system support.	Time-critical for extensive burns and severe cutaneous drug reactions.	Fluid, infection, scarring, organ injury and prolonged rehabilitation.	Physiology, wound closure, infection, organ function, pain and participation.
Behavioral and itch/pain interventions	Whole-person	Reduce scratching/picking, improve sleep and coping, and treat comorbid mental-health mechanisms.	Useful as adjunct or primary treatment for defined body-focused conditions.	Stigma and misuse to dismiss biological disease.	Skin injury, behavior frequency, sleep, distress and function.
Cosmetic and appearance-related intervention	Elective/psych	Microdermabrasion, hair, scar, contour or texture.	Can improve selected patient-valued outcomes; evidence and regulation vary widely.	Burns/scars, pigment injury, unrealistic expectations, financial harm and body-image pathology.	Informed goals, standardized imaging, adverse effects, satisfaction and mental-health screening when indicated.
Public-health and occupational skin programs	Prevention/system	PPE, safer products, sanitation, early treatment and access for infectious/occupational/neglected skin disease.	High impact where exposures and care gaps are structural.	PPE dermatitis, stigma and program sustainability.	Exposure, incidence, treatment completion, work participation and equity.

9.0.1 Barrier care and irritant reduction

Target and mechanism. Restore moisture/lipids, reduce friction/wet work and avoid damaging exposures.

Evidence and expected outcome. Foundational for dermatitis, xerosis and wound prevention.

Risks, burden, and equity. Cost, contact allergy, adherence and occupational infeasibility.

Response monitoring. Symptoms, barrier measures, flares, skin integrity and exposure.

Evidence anchors: [3,14]

9.0.2 Topical pharmacotherapy

Target and mechanism. Deliver anti-inflammatory, antimicrobial, keratolytic, retinoid or other agents to skin.

Evidence and expected outcome. Strong for many localized conditions when vehicle and adherence fit.

Risks, burden, and equity. Irritation, atrophy/systemic absorption, resistance and contact allergy.

Response monitoring. Lesion response, adverse effects, amount/use and escalation need.

Evidence anchors: [3,10]

9.0.3 Photoprotection and therapeutic phototherapy

Target and mechanism. Reduce harmful UV exposure or deliver controlled wavelengths for selected inflammatory/pigment disease.

Evidence and expected outcome. Sun protection reduces UV injury; phototherapy effective for selected diseases.

Risks, burden, and equity. Vitamin/behavior concerns are often overstated; burns, photoaging and logistical burden.

Response monitoring. Exposure, lesion response, cumulative dose and adverse effects.

Evidence anchors: [3,9]

9.0.4 Systemic and biologic dermatologic therapy

Target and mechanism. Modulate immune, hormonal, microbial or keratinization pathways.

Evidence and expected outcome. Major benefit in selected severe psoriasis, dermatitis, acne and other disorders.

Risks, burden, and equity. Infection, organ/reproductive effects, monitoring and cost.

Response monitoring. Disease target, labs/organ safety, infection, QOL and adherence.

Evidence anchors: [3]

9.0.5 Antimicrobial stewardship and source control

Target and mechanism. Use appropriate topical/systemic agents, drainage/debridement and hygiene.

Evidence and expected outcome. Effective when infection is correctly identified and source controlled.

Risks, burden, and equity. Resistance, allergy, microbiome injury and unnecessary treatment.

Response monitoring. Culture/susceptibility, response, recurrence and antimicrobial use.

Evidence anchors: [10,11]

9.0.6 Wound offloading, compression and perfusion treatment

Target and mechanism. Remove pressure/shear, control edema and restore blood flow where possible.

Evidence and expected outcome. Often more important than advanced dressings for chronic wounds.

Risks, burden, and equity. Ischemia risk with compression, adherence, device injury and access.

Response monitoring. Pressure/load, perfusion, edema, wound trajectory, infection and function.

Evidence anchors: [2,12]

9.0.7 Surgery and procedural dermatology

Target and mechanism. Biopsy, excision, Mohs surgery, drainage, laser or reconstructive procedures.

Evidence and expected outcome. Strong for many tumors and selected structural/inflammatory problems.

Risks, burden, and equity. Scar, infection, pigment change, recurrence and access.

Response monitoring. Margins/pathology, healing, recurrence, function and cosmetic goals.

Evidence anchors: [3,9]

9.0.8 Emergency burn and severe-reaction care

Target and mechanism. Stabilize airway/circulation, stop exposure, control temperature/pain, remove culprit and provide specialized wound/system support.

Evidence and expected outcome. Time-critical for extensive burns and severe cutaneous drug reactions.

Risks, burden, and equity. Fluid, infection, scarring, organ injury and prolonged rehabilitation.

Response monitoring. Physiology, wound closure, infection, organ function, pain and participation.

Evidence anchors: [12,30]

9.0.9 Behavioral and itch/pain interventions

Target and mechanism. Reduce scratching/picking, improve sleep and coping, and treat comorbid mental-health mechanisms.

Evidence and expected outcome. Useful as adjunct or primary treatment for defined body-focused conditions.

Risks, burden, and equity. Stigma and misuse to dismiss biological disease.

Response monitoring. Skin injury, behavior frequency, sleep, distress and function.

Evidence anchors: [21,32]

9.0.10 Cosmetic and appearance-related intervention

Target and mechanism. Alter pigmentation, hair, scar, contour or texture.

Evidence and expected outcome. Can improve selected patient-valued outcomes; evidence and regulation vary widely.

Risks, burden, and equity. Burns/scars, pigment injury, unrealistic expectations, financial harm and body-image pathology.

Response monitoring. Informed goals, standardized imaging, adverse effects, satisfaction and mental-health screening when indicated.

Evidence anchors: [30,31]

9.0.11 Public-health and occupational skin programs

Target and mechanism. PPE, safer products, sanitation, early treatment and access for infectious/occupational/neglected skin disease.

Evidence and expected outcome. High impact where exposures and care gaps are structural.

Risks, burden, and equity. PPE dermatitis, stigma and program sustainability.

Response monitoring. Exposure, incidence, treatment completion, work participation and equity.

Evidence anchors: [4,14,33]

What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
Can images safely triage skin disease across all skin tones?	Expert imaging and dermoscopy support many diagnostic pathways.	Transportable autonomous performance and net clinical benefit across devices, sites and populations.	Training imbalance, spectrum/lighting shift and weak labels create shortcuts.	Prospective diverse multicenter trials with pathology/outcome reference and failure abstention.
Which wound biomarkers predict healing or infection?	Temperature, oxygen, pH and protease/microbial signals correlate with aspects of wound state.	Incremental utility beyond cause-focused assessment and serial geometry.	Biofilm, exudate, sampling and interventions change signals.	Locked sensor studies tied to reference perfusion, microbiology and healing decisions.
How much risk is genetic versus environmental?	Both inherited variation and environments contribute, often through interaction and mediation across the life course.	No single percentage applies to all outcomes, populations, ages, or definitions within integumentary and dermatologic health.	Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.	Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.
Can one composite score summarize health?	Composite scores can rank a defined outcome in a defined population and may improve workflow.	Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.	Different variables compensate, interact, and have different treatment implications; weights drift over time.	Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

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Question	What is established	What remains uncertain	Why unresolved	Decisive next step
When is earlier detection beneficial?	Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.	Net benefit for many incidental, molecular, wearable, and multi-cancer signals.	Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.	Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.
Do digital biomarkers generalize?	Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.	Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.	Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.	Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.
What counts as meaningful change?	Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.	Thresholds for many biomarkers, sensors, and multidomain profiles.	Baseline severity, goals, adaptation, and treatment burden alter value.	Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

10.0.1 Can images safely triage skin disease across all skin tones?

Established knowledge. Expert imaging and dermoscopy support many diagnostic pathways.

Unresolved issue. Transportable autonomous performance and net clinical benefit across devices, sites and populations.

Why current evidence is insufficient. Training imbalance, spectrum/lighting shift and weak labels create shortcuts.

Research that would change confidence. Prospective diverse multicenter trials with pathology/outcome reference and failure abstention.

Evidence anchors: ^[15,29]

10.0.2 Which wound biomarkers predict healing or infection?

Established knowledge. Temperature, oxygen, pH and protease/microbial signals correlate with aspects of wound state.

Unresolved issue. Incremental utility beyond cause-focused assessment and serial geometry.

Why current evidence is insufficient. Biofilm, exudate, sampling and interventions change signals.

Research that would change confidence. Locked sensor studies tied to reference perfusion, microbiology and healing decisions.

Evidence anchors: [23,26]

10.0.3 How much risk is genetic versus environmental?

Established knowledge. Both inherited variation and environments contribute, often through interaction and mediation across the life course.

Unresolved issue. No single percentage applies to all outcomes, populations, ages, or definitions within integumentary and dermatologic health.

Why current evidence is insufficient. Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

Research that would change confidence. Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

10.0.4 Can one composite score summarize health?

Established knowledge. Composite scores can rank a defined outcome in a defined population and may improve workflow.

Unresolved issue. Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

Why current evidence is insufficient. Different variables compensate, interact, and have different treatment implications; weights drift over time.

Research that would change confidence. Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Evidence anchors: [29,32]

10.0.5 When is earlier detection beneficial?

Established knowledge. Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

Unresolved issue. Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

Why current evidence is insufficient. Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

Research that would change confidence. Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

Evidence anchors: ^[9]

10.0.6 Do digital biomarkers generalize?

Established knowledge. Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

Unresolved issue. Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

Why current evidence is insufficient. Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

Research that would change confidence. Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

Evidence anchors: ^[23]

10.0.7 What counts as meaningful change?

Established knowledge. Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

Unresolved issue. Thresholds for many biomarkers, sensors, and multidomain profiles.

Why current evidence is insufficient. Baseline severity, goals, adaptation, and treatment burden alter value.

Research that would change confidence. Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

Evidence anchors: ^[32]

Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
0. Skin-tone-calibrated optical atlas	How do reflectance and lesion features vary by wavelength, site, physiology and skin characteristics?	Diverse prospective cohort with calibrated multispectral/dermoscopic images, pathology where indicated and longitudinal outcomes.	Open reference distributions and robust externally validated algorithms.	Performance remains device/site-specific or excludes darker skin.
0. Smart wound utility trial	Does continuous pressure/temperature/moisture sensing reduce ulcer progression or recurrence?	Bench and biocompatibility verification followed by pragmatic randomized workflow trial.	Lower patient-important wound burden without alarm or skin-contact harm.	More data and interventions without improved healing or recurrence.
1. Reference map	What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for integumentary and dermatologic health?	Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.	Reproducible distributions and state-dependent reference models with quantified uncertainty.	Convenience sample presented as universal normality.
2. Causal determinants	Which modifiable exposures cause clinically and functionally important outcomes, and in whom?	Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.	Convergent effect estimates linked to plausible mechanisms and actionable interventions.	Association-only result marketed as prevention guidance.
3. Measurement validity	Does the new device or biomarker measure the intended quantity under real use?	Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.	Prespecified error and failure limits met across intended-use conditions.	High average correlation with systematic individual error or unreported exclusions.

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Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
4. Clinical utility	Does using the measurement improve decisions or outcomes compared with current care?	Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.	Net benefit on patient-important outcomes with acceptable implementation burden.	Improved prediction without changed decisions or outcomes.
5. Longitudinal multimorbidity	How does the domain interact with other conditions, treatments, and functional trajectories?	Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.	Models that support coordinated care rather than isolated disease optimization.	Single-disease endpoint worsens overall burden or conflicts with goals.
6. Implementation and governance	Can effective interventions be delivered safely, equitably, affordably, and sustainably?	Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.	Sustained adoption with monitored outcomes and reduced inequity.	Pilot success depends on exceptional staffing or hidden externalized costs.

11.0.1 0. Skin-tone-calibrated optical atlas

Question. How do reflectance and lesion features vary by wavelength, site, physiology and skin characteristics?

Design. Diverse prospective cohort with calibrated multispectral/dermoscopic images, pathology where indicated and longitudinal outcomes.

Success criterion. Open reference distributions and robust externally validated algorithms.

Failure or stop criterion. Performance remains device/site-specific or excludes darker skin.

Evidence anchors: ^[15]

11.0.2 0. Smart wound utility trial

Question. Does continuous pressure/temperature/moisture sensing reduce ulcer progression or recurrence?

Design. Bench and biocompatibility verification followed by pragmatic randomized workflow trial.

Success criterion. Lower patient-important wound burden without alarm or skin-contact harm.

Failure or stop criterion. More data and interventions without improved healing or recurrence.

Evidence anchors: ^[12,23]

11.0.3 1. Reference map

Question. What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for integumentary and dermatologic health?

Design. Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

Success criterion. Reproducible distributions and state-dependent reference models with quantified uncertainty.

Failure or stop criterion. Convenience sample presented as universal normality.

Evidence anchors: ^[15]

11.0.4 2. Causal determinants

Question. Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

Design. Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

Success criterion. Convergent effect estimates linked to plausible mechanisms and actionable interventions.

Failure or stop criterion. Association-only result marketed as prevention guidance.

11.0.5 3. Measurement validity

Question. Does the new device or biomarker measure the intended quantity under real use?

Design. Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

Success criterion. Prespecified error and failure limits met across intended-use conditions.

Failure or stop criterion. High average correlation with systematic individual error or unreported exclusions.

Evidence anchors: ^[26,29]

11.0.6 4. Clinical utility

Question. Does using the measurement improve decisions or outcomes compared with current care?

Design. Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

Success criterion. Net benefit on patient-important outcomes with acceptable implementation burden.

Failure or stop criterion. Improved prediction without changed decisions or outcomes.

Evidence anchors: ^[33]

11.0.7 5. Longitudinal multimorbidity

Question. How does the domain interact with other conditions, treatments, and functional trajectories?

Design. Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

Success criterion. Models that support coordinated care rather than isolated disease optimization.

Failure or stop criterion. Single-disease endpoint worsens overall burden or conflicts with goals.

11.0.8 6. Implementation and governance

Question. Can effective interventions be delivered safely, equitably, affordably, and sustainably?

Design. Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

Success criterion. Sustained adoption with monitored outcomes and reduced inequity.

Failure or stop criterion. Pilot success depends on exceptional staffing or hidden externalized costs.

Evidence anchors: ^[15,33]

Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

Volume	Companion topic	Relationship	Use this link when...
00	Atlas of Human Health	Master taxonomy, history of classification, and multi-axial architecture.	locating the primary home of a concept or mapping classifications.
13	Measurement, Evidence, Data, and Informatics	Shared metrology, causal inference, validation, interoperability, and data governance.	designing a study, sensor, dataset, model, or evidence claim.
14	Interventions and Therapeutics	Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety.	moving from measurement to action.
15	Ethics, Law, Rights, Equity, and Governance	Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.	people, identifiable data, invasive procedures, or consequential algorithms are involved.
03H	Immune and Lymphatic Health	Barrier immunity, allergy, autoimmunity and lymphatic edema are shared mechanisms.	rash or wound phenotype is immune-mediated or recurrent.
09	Environmental and Occupational Health	UV, heat/cold, chemicals, allergens, wet work and infection exposures shape disease.	causal prevention depends on exposure.
03D	Endocrine and Metabolic Health	Diabetes, hormones, nutrition and metabolism alter skin, hair and healing.	wounds, acne, pigment or appendage changes suggest systemic coupling.
05	Brain, Cognition, Emotion and Behavior	Itch, pain, picking, appearance and stress influence symptoms and care.	skin and behavior form a feedback loop.
10	Function, Disability and Rehabilitation	Burns, scars, wounds, lymphedema and assistive-device interfaces affect participation.	healing must translate to function.

Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

Minimum data schema for a longitudinal domain profile

Layer	Minimum fields	Critical caution
Identity and context	Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context.	Avoid collecting identity variables without purpose; preserve privacy and provenance.
Time	Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.	Clock drift and ambiguous time zones can invalidate multimodal inference.
Raw signal / specimen	Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.	Derived scores without raw or quality data are difficult to audit.
Quality	Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.	Quality must be a first-class variable, not only an exclusion after analysis.
Reference	Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.	A weak reference caps the validity of the new system.
Phenotype and function	Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.	Disease codes alone do not represent lived outcome.
Intervention and response	Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.	Confounding by indication and treatment changes must be modeled.
Governance	Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.	Data stewardship is part of scientific validity.

Glossary

Term	Operational meaning in this volume
Transepidermal water loss	Water-vapor flux from body through skin; a context-sensitive proxy of barrier function.
Dermoscopy	Magnified optical examination that reveals subsurface pigment and vascular structures.
Biofilm	Structured microbial community attached to a surface and embedded in matrix, often more tolerant of treatment.
Erythema	Visible redness usually related to vascular change; optical appearance varies with skin pigmentation and lighting.
Allostasis	Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.
Biomarker	Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint.
Function	What a person can do in body functions, activities, and participation, interpreted in context.
Homeostasis	Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.
Phenotype	Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.
Risk factor	Characteristic associated with future outcome; causal status and modifiability must be established separately.
Calibration	Establishing the relationship between an instrument output and a reference quantity under specified conditions.
Clinical validity	Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.
Intended use	The specific purpose, population, setting, user, input, and decision for which a device or model is designed.
Reference method	Comparator selected to estimate truth for validation; every reference has its own error and scope.
Reserve	Capacity to maintain function under challenge beyond resting baseline.
Trajectory	Direction, rate, variability, and reversibility of change over time.

Selected references and current standards

The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

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Version note

Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.